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TIME-Net: Transformer-Integrated Multi-Encoder Network for limited-angle artifact removal in dual-energy CBCT

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ABSTRACT

Dual-energy cone-beam computed tomography (DE-CBCT) is a promising imaging technique with foreseeable clinical applications. DE-CBCT images acquired with two different spectra can provide material-specific information. Meanwhile, the anatomical consistency and energy-domain correlation result in significant information redundancy, which could be exploited to improve image quality. In this context, this paper develops the Transformer-Integrated Multi-Encoder Network (TIME-Net) for DE-CBCT to remove the limited-angle artifacts. TIME-Net comprises three encoders (image encoder, prior encoder, and transformer encoder), two decoders (low- and high-energy decoders), and one feature fusion module. Three encoders extract various features for image restoration. The feature fusion module compresses these features into more compact shared features and feeds them to the decoders. Two decoders perform differential learning for DE-CBCT images. By design, TIME-Net could obtain high-quality DE-CBCT images using two complementary quarter-scans, holding great potential to reduce radiation dose and shorten the acquisition time. Qualitative and quantitative analyses based on simulated data and real rat data have demonstrated the promising performance of TIME-Net in artifact removal, subtle structure restoration, and reconstruction accuracy preservation. Two clinical applications, virtual non-contrast (VNC) imaging and iodine quantification, have proved the potential utility of the DE-CBCT images provided by TIME-Net.

1. Introduction

Dual-energy computed tomography (DECT) is an emerging quantitative imaging technique with many clinical applications, such as virtual monochromatic imaging (Pomerantz et al., 2013), perfused blood volume (Meinel et al., 2014), automated bone removal (Schulz et al., 2012), urinary stone characterization (Leng et al., 2015), etc. And the interest in applying the dual-energy technique to optimize radiotherapy also is increasing as DECT imaging has gained widespread acceptance in the diagnostic radiology community (Van Elmpt et al., 2016). On the other hand, cone-beam computed tomography (CBCT) is an effective imaging modality using the flat-panel detector and has been widely utilized in clinical practice, including dentistry (Kiljunen et al., 2015), radiotherapy (Jaffray et al., 2002) and interventions (Wallace et al., 2008). However, the x-ray scattering and beam hardening may lead to image inhomogeneity, compromising the visibility of soft tissues and the accuracy of CT numbers.

Dual-energy imaging technology brings new opportunities to CBCT (Sajja et al., 2020; Schyns et al., 2017). DE-CBCT not only can improve soft tissue contrast but also have the ability of quantitative imaging. Researchers hold a continuing interest in clinical applications of DE-CBCT, although it has not yet become routine clinical practice. These efforts involve fields such as dentistry (Kim et al., 2019; Chang and Lee, 2021; Li et al., 2022), interventions (Müller et al., 2016), and radiotherapy (Hoggarth et al., 2013; Sherertz et al., 2014; Men et al., 2017; Skaarup et al., 2019; Cassetta et al., 2020; Jiang et al., 2021). The promising results presented in the preliminary studies mentioned above are encouraging the development of algorithms associated with DE-CBCT.

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Compared to single-energy CT, the dual-energy imaging technique allows the identification and quantification of different materials through an additional measurement acquired with a second x-ray spectrum (Mccollough et al., 2015). The anatomical consistency and energy-domain correlation result in significant information redundancy, which has been widely explored to improve image quality, reduce radiation dose or relax imaging conditions. A classical scheme is to design various regularization terms using this kind of redundancy for iterative reconstruction. For example, the average image-induced non-local means filter, which exploits the energy domain redundancy, was developed to reduce the noise-induced artifacts in DECT images (Zhang et al., 2017). Zhang et al. (2020b) and Chen et al. (2018a) reconstructed DECT or material images from two partial-angular scans using prior knowledge and iterative algorithms. Recently, Chen et al. (2021a) introduced the directional-total-variation to remove the limited-angle artifacts for DECT. Even though these methods have demonstrated promising results, the involved hyper-parameters need to be adjusted empirically, and the whole reconstruction process usually is time-consuming. Moreover, most researchers do not have access to the raw data of commercial scanners, which are necessary for data fidelity in iterative reconstruction (Shan et al., 2019). These limitations heavily hinder the deployment of iterative reconstruction methods in clinical practice.

In recent years, deep learning (DL) has become a new frontier in the medical imaging community (Zhu et al., 2018; Wang et al., 2018a, 2020). The promising results of deep learning in single-energy CT applications (Yin et al., 2019; Chen et al., 2018b; Zhang et al., 2021a; Chen et al., 2017; Yang et al., 2018) have encouraged the development of DL-based DECT algorithms (Zhang et al., 2019b, 2020c, 2021b; Lyu et al., 2021; Wang et al., 2021b; Zhang et al., 2020a). However, the limited receptive field of convolution operation makes it difficult for convolutional neural networks (CNNs) to explore the similarity across large regions (Wang et al., 2018b; Zhang et al., 2019a), compromising the capability of CNN-based algorithms for modeling various structural features (Li et al., 2020). Recently, transformers, which can capture global knowledge by modeling long-range dependency, have emerged as a powerful tool for many tasks, such as natural language processing (Wolf et al., 2020), object detection (Carion et al., 2020), and image segmentation (Liu et al., 2021). Some efforts have also demonstrated its promising performance in medical image segmentation (Heinrich et al., 2019, 2021; Dong et al., 2020), magnetic resonance imaging (Feng et al., 2021), and CT denoising (Zhang et al., 2021c). What is more, compared to iterative reconstruction, the DL-based post-processing strategies have demonstrated broad applicability because these methods do not rely on raw data (Shan et al., 2019).

In order to exploit the redundant information using deep learning, this paper follows previous studies (Zhang et al., 2021b; Wang et al., 2022; Zhang et al., 2020a) and suggests acquiring dual-energy attenuation measurements using two complementary scans, as illustrated in Fig. 1. When the scanning angle α equals 90°, we can get two complementary quarter-scans, as indicated by the trajectory t2 in Fig. 1. It can be observed from Fig. 1(b) and (c) that the raw DE-CBCT limited-angle images suffer from severe wedge artifacts. While the fused prior image only suffers from slight artifacts and retains a lot of image details, which can be used to improve DE-CBCT imaging.

Consequently, this paper develops the Transformer-Integrated Multi-Encoder Network (TIME-Net) for DE-CBCT. TIME-Net exploits various information through three different encoders to improve imaging quality. First, the image encoder extracts high-level image features using convolutional neural networks. Second, another CNN-based prior encoder captures structural information by making full use of the redundant information. Third, the transformer encoder takes the global knowledge to improve the representation capacity of TIME-Net. Then, various features extracted by the three encoders are fused and compressed as shared features with the fusion module. Finally, the low- and high-energy decoders perform differential learning to produce high-quality DE-CBCT images from the shared features. To evaluate the proposed algorithm, both the simulated data and real rat data are utilized in experiments.

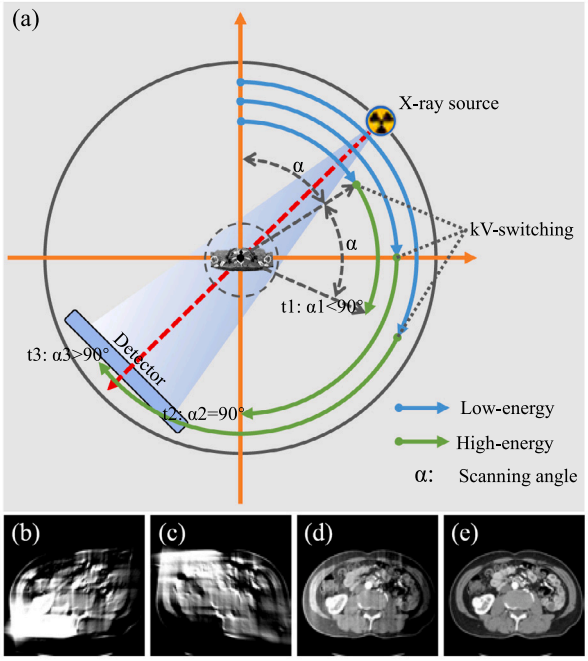


Fig. 1. (a) is the complementary scan configuration. (b) and (c) are the limited-angle DE-CBCT images. (d) is the prior image fused using (b) and (c). (e) is the reference low-energy image.

2. Material and methods

2.1. Limited-angle artifact removal via TIME-Net

In view of the broad applicability of post-processing strategies, this paper is dedicated to finding an image domain network $\boldsymbol{\Phi}^*_{\mathrm{ID}}$ that can accurately estimate the limited-angle artifacts from the degraded DE-CBCT images, which can be defined as:

$$\begin{pmatrix}\mathbf{I}^{\mathrm{L}}_{\mathrm{LA}}\\ \mathbf{I}^{\mathrm{H}}_{\mathrm{LA}}\end{pmatrix} - \boldsymbol{\Phi}^*_{\mathrm{ID}}\left(\begin{pmatrix}\mathbf{I}^{\mathrm{L}}_{\mathrm{LA}}\\ \mathbf{I}^{\mathrm{H}}_{\mathrm{LA}}\end{pmatrix}, \mathbf{I}_{\mathrm{Prior}}\right) \approx \begin{pmatrix}\mathbf{I}^{\mathrm{L}}_{\mathrm{Ref}}\\ \mathbf{I}^{\mathrm{H}}_{\mathrm{Ref}}\end{pmatrix}, \tag{1}$$

where

$$\boldsymbol{\Phi}^*_{\mathrm{ID}} = \arg\min_{\boldsymbol{\Phi}_{\mathrm{ID}}} \left\| \boldsymbol{\Phi}_{\mathrm{ID}}\left(\begin{pmatrix}\mathbf{I}^{\mathrm{L}}_{\mathrm{LA}}\\ \mathbf{I}^{\mathrm{H}}_{\mathrm{LA}}\end{pmatrix}, \mathbf{I}_{\mathrm{Prior}}\right) - \begin{pmatrix}\mathbf{I}^{\mathrm{L}}_{A}\\ \mathbf{I}^{\mathrm{H}}_{A}\end{pmatrix} \right\|, \tag{2}$$

$$\begin{pmatrix}\mathbf{I}^{\mathrm{L}}_{\mathrm{LA}}\\ \mathbf{I}^{\mathrm{H}}_{\mathrm{LA}}\end{pmatrix} = \mathrm{FDK}\begin{pmatrix}\mathbf{P}^{\mathrm{L}}_{\mathrm{LA}}\\ \mathbf{P}^{\mathrm{H}}_{\mathrm{LA}}\end{pmatrix}, \tag{3}$$

$$\mathbf{I}_{\mathrm{Prior}} = \frac{1}{2}\cdot\mathbf{I}^{\mathrm{L}}_{\mathrm{LA}} + \frac{1}{2}\cdot\mathbf{I}^{\mathrm{H}}_{\mathrm{LA}}. \tag{4}$$

$\mathbf{I}^{\mathrm{E}}_{\mathrm{LA}}$, $\mathbf{I}^{\mathrm{E}}_{\mathrm{Ref}}$, $\mathbf{I}^{\mathrm{E}}_{\mathrm{A}}$, and $\mathbf{P}^{\mathrm{E}}_{\mathrm{LA}}$ with $\mathrm{E} \in \{\mathrm{L},\mathrm{H}\}$ represent the limited-angle image, the reference image, the artifact image, and the limited-angle projection at low- and high-energy, respectively. $\mathbf{I}_{\mathrm{Prior}}$ denotes the prior image fused using the limited-angle DE-CBCT images. Please note that $\mathbf{I}_{\mathrm{Prior}}$ also can be reconstructed through the FDK algorithm using the fused DE projection data, but the adopted average operation is more efficient than FDK. The well-trained TIME-Net is a specific case of $\boldsymbol{\Phi}^*_{\mathrm{ID}}$. FDK is the analytical volume reconstruction algorithm.

2.2. TIME-Net

TIME-Net is developed for solving the optimization problem formulated in Eq. (2). Overall, TIME-Net takes the prior image and degraded DE-CBCT images as input and outputs an estimation of the limited-angle artifacts. Then the high-quality DE-CBCT images can be obtained by eliminating the artifact components from the degraded DE-CBCT

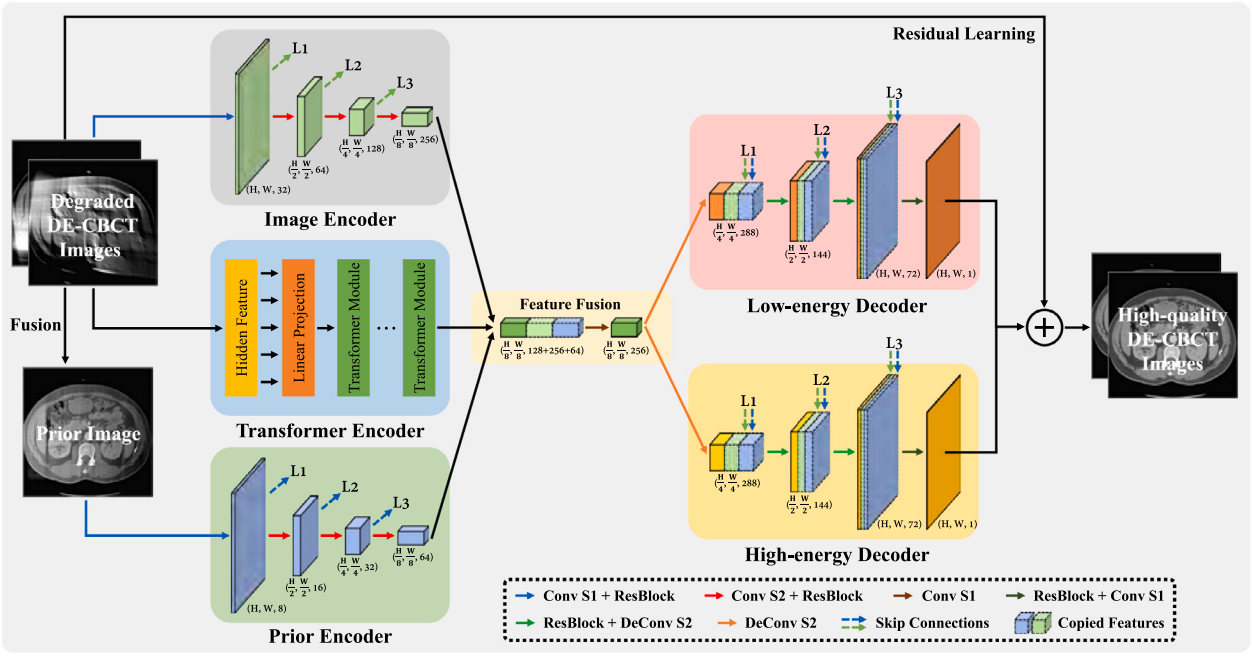


Fig. 2. The schematic of the proposed TIME-Net. On the whole, TIME-Net comprises three encoders (Image Encoder, Transformer Encoder, and Prior Encoder), two decoders (Low- and High-energy Encoder), and one feature fusion module.

images. As shown in Fig. 2, TIME-Net comprises three encoders, two decoders, and one feature fusion module, with each one being responsible for a specific task. First, the CNN-based image encoder extracts high-level image features from the degraded DE-CBCT images. Second, the prior encoder uses the prior image to capture the structural information caused by anatomical consistency. Third, the transformer encoder, which could exploit global knowledge, is integrated into the TIME-Net to enhance its representation capacity. Then, the various features are fused and compressed as shared features through the feature fusion module. Finally, to capture the differences between DE-CBCT images, the low- and high-energy decoders are utilized for differential learning.

2.2.1. Image encoder and prior encoder

As illustrated in Fig. 2, the image encoder and prior encoder have the same structure but different inputs. The image encoder takes degraded DE-CBCT images as input directly, while the prior encoder takes the prior image as input. Besides, as the image encoder is the primary module, its initial channel number is set to 32. While the prior encoder as an auxiliary component is assigned fewer parameters, and its initial channel number is set to 8. Both the image encoder and prior encoder are modified from U-Net (Ronneberger et al., 2015). U-Net and its variants have demonstrated promising performance in various medical applications, such as segmentation (Wu et al., 2021), denoising (Gu et al., 2021), and image synthesis (Xu et al., 2020). Here we simplify its structure to make it more efficient, and the cascaded convolutional layers are replaced with residual blocks to tackle the gradient vanishing problem. Given degraded DE-CBCT images $\begin{pmatrix}\mathbf{I}^{\mathrm{L}}_{\mathrm{LA}}\\ \mathbf{I}^{\mathrm{H}}_{\mathrm{LA}}\end{pmatrix} \in \mathbb{R}^{\mathrm{W}\times\mathrm{H}\times 2}$ and the prior image $\mathbf{I}_{\mathrm{Prior}} \in \mathbb{R}^{\mathrm{W}\times\mathrm{H}\times 1}$ with the spatial resolution is $\mathrm{W}\times\mathrm{H}$, the features extracted by the image encoder and prior encoder are $\mathbf{F}_{\mathrm{IE}} \in \mathbb{R}^{\frac{\mathrm{W}}{8}\times\frac{\mathrm{H}}{8}\times 256}$ and $\mathbf{F}_{\mathrm{PE}} \in \mathbb{R}^{\frac{\mathrm{W}}{8}\times\frac{\mathrm{H}}{8}\times 64}$, respectively.

2.2.2. Transformer encoder

Following previous studies (Chen et al., 2021b; Zhang et al., 2021c; Dosovitskiy et al., 2020), the degraded DE-CBCT images $\begin{pmatrix}\mathbf{I}^{\mathrm{L}}_{\mathrm{LA}}\\ \mathbf{I}^{\mathrm{H}}_{\mathrm{LA}}\end{pmatrix}$ are first serialized into a sequence of image patches $\left\{\mathbf{x}^{\mathrm{n}}_{\mathrm{p}} \in \mathbb{R}^{\mathrm{P}\times\mathrm{P}\times 2} | \mathrm{n}=1,\ldots,\mathrm{N}\right\}$, where each patch is of size $\mathrm{P}\times\mathrm{P}\times 2$ and $\mathrm{N}=\frac{\mathrm{WH}}{\mathrm{P}^2}$ is the number of image patches. Then a trainable linear projection $\mathbf{E} \in \mathbb{R}^{2\mathrm{P}^2\times D}$ is used to map the vectorized image patches into the latent D-dimensional embedding space $\left[\mathbf{x}^1_{\mathrm{p}}\mathbf{E}; \mathbf{x}^2_{\mathrm{p}}\mathbf{E}, \ldots, \mathbf{x}^{\mathrm{N}}_{\mathrm{p}}\mathbf{E}\right]$. Furthermore, to explore the spatial information among patches, a specific trainable position embedding $\mathbf{E}_{\mathrm{pos}} \in \mathbb{R}^{\mathrm{N}\times D}$ is added to the patch embeddings. Then the intermediate embeddings $\mathbf{E}_{\mathrm{inter}} = \left[\mathbf{x}^1_{\mathrm{p}}\mathbf{E}; \mathbf{x}^2_{\mathrm{p}}\mathbf{E}, \ldots, \mathbf{x}^{\mathrm{N}}_{\mathrm{p}}\mathbf{E}\right] + \mathbf{E}_{\mathrm{pos}}$ are fed into multiple cascaded transformer modules for further encoding. As illustrated in Fig. 3, each transformer module consists of three kinds of cascaded components, i.e., Layer Normalization (LN) operator, Multi-Head Self-Attention (MHSA) block, and Multi-Layer Perceptron (MLP) block. Finally, the product of the transformer encoder is the coded image representation $\mathbf{F}_{\mathrm{TE}} \in \mathbb{R}^{\frac{\mathrm{W}}{\mathrm{P}}\times\frac{\mathrm{H}}{\mathrm{P}}\times D}$. To match the shape of features extracted by the other two encoders, the size of image patches is set to $8\times 8\times 2$ (i.e., $\mathrm{P}=8$). Furthermore, after considering the trade-off between model performance and computational burden, the dimension of latent embedding space D and the number of transformer modules were set to 128 and 3, respectively.

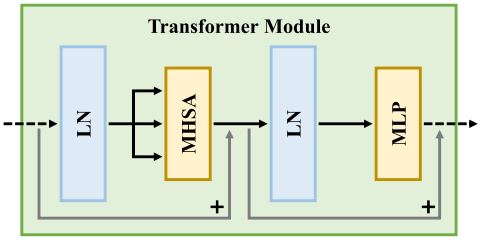


Fig. 3. The schematic of transformer module. LN: Layer Normalization operator, MHSA: Multi-Head Self-Attention block. MLP: Multi-Layer Perceptron block.

2.2.3. Feature fusion module

As illustrated in Fig. 2, a feature fusion module is employed to fuse and compress various features. First, all the features are concatenated as the intermediate features $\left[\mathbf{F}_{\mathrm{IE}}, \mathbf{F}_{\mathrm{PE}}, \mathbf{F}_{\mathrm{TE}}\right] \in \mathbb{R}^{\frac{\mathrm{W}}{8}\times\frac{\mathrm{H}}{8}\times(256+64+128)}$. And then, the intermediate features are compressed into the shared features $\mathbf{F}_{\mathrm{shared}} \in \mathbb{R}^{\frac{\mathrm{W}}{8}\times\frac{\mathrm{H}}{8}\times 256}$ by a convolutional layer.

2.2.4. Low- and high-energy decoders

Differences between DE-CBCT images are the key dependencies for material identification and quantification. Therefore, TIME-Net adopts two decoders (each with one output channel) rather than one decoder (with two output channels) to deal with the dual-energy data because the former scheme holds more trainable parameters for differentiated learning. The low- and high-energy decoders are of the same structure and modified based on U-Net as well. During decoding, the shallow feature maps of the image encoder and prior encoder are copied and concatenated (the green and blue dashed arrows) to the middle layers of low- and high-energy decoders, as shown in Fig. 2. The two decoders can estimate the limited-angle artifact components $\binom{\mathbf{I}_{\mathrm{A}}^{\mathrm{L}}}{\mathbf{I}_{\mathrm{A}}^{\mathrm{H}}} \in \mathbb{R}^{\mathrm{W}\times\mathrm{H}\times 2}$ contained in the degraded DE-CBCT images from the coded various features.

2.2.5. Loss function

Previous studies have demonstrated that an appropriate loss function helps to improve the performance of the deep learning model. Therefore, in this study, a composite loss function using Mean Absolute Error (MAE) and Structural Similarity (SSIM) metrics is designed for TIME-Net, as formulated in Eq. (5).

$$\begin{aligned}\mathcal{L}_{\mathrm{com}} = \sigma\cdot\mathcal{L}_{\mathrm{MAE}}\left(\binom{\mathbf{I}_{\mathrm{Ref}}^{\mathrm{L}}}{\mathbf{I}_{\mathrm{Ref}}^{\mathrm{H}}},\binom{\mathbf{I}_{\mathrm{Pred}}^{\mathrm{L}}}{\mathbf{I}_{\mathrm{Pred}}^{\mathrm{H}}}\right)+\\ \beta\cdot\mathcal{L}_{\mathrm{SSIM}}\left(\binom{\mathbf{I}_{\mathrm{Ref}}^{\mathrm{L}}}{\mathbf{I}_{\mathrm{Ref}}^{\mathrm{H}}},\binom{\mathbf{I}_{\mathrm{Pred}}^{\mathrm{L}}}{\mathbf{I}_{\mathrm{Pred}}^{\mathrm{H}}}\right)+\\ \gamma\cdot\mathcal{L}_{\mathrm{MAE}}\left(\mathbf{D}_{\mathrm{Ref}},\mathbf{D}_{\mathrm{Pred}}\right).\end{aligned} \tag{5}$$

where $\mathbf{D}$ denotes the differences between low- and high-energy data. σ, β and γ are the weights balancing different loss terms. In this study, these three hyper-parameters are set as $\sigma = 1$, $\beta = 100$ and $\gamma = 0.5$ based on extensive preliminary experiments.

2.3. Dataset

2.3.1. Simulated data

The unavailability of raw data to researchers is a long-standing problem in the medical imaging community (Yang et al., 2018; Shan et al., 2019; Zhang et al., 2021a). To address this challenge, this study utilized the clinical DECT images acquired from 25 patients who had iodine contrast-enhanced DECT examinations to generate a simulated DE-CBCT dataset. These scans were performed in Nanjing General PLA Hospital, China, with the approval of the institutional review board and patient consent forms. After administering the iodine contrast agent, the DECT images were acquired using a SOMATOM Definition Flash DECT scanner (Siemens Healthineers, Germany). The low- and high-energy of DECT scans were 100 kVp and 140 kVp, respectively. The dataset was randomly split into a training set (21 patients, 8000 slices), a validation set (2 patients, 400 slices), and a testing set (2 patients, 607 slices). A cone-beam imaging geometry was adopted. The distances of the source to detector and the source to object were 1500.0 mm and 1000.0 mm, respectively. The detector had 900 × 400 units, with each covering a region of 1.5×1.5 mm^2. The reconstructed volume had 512 × 512 × 200 voxels, and each voxel covered a space of $0.9 \times 0.9 \times 0.9$ mm^3. The full-scan had 960 views acquired uniformly over a 360-degree range. The measurements from 0° to 90° in the low-energy full-scan were extracted as the low-energy quarter-scan, and the measurements from 90° to 180° in the high-energy full-scan were taken as the high-energy quarter-scan.

2.3.2. Real rat data

The real DE-CBCT data acquired with some rats also were used to evaluate the proposed algorithm. These scans were performed on an in-house Micro-CT imaging system with a microfocus x-ray tube (L9421-02, Hamamatsu Photonics, Hamamatsu, Japan) and a flat-panel detector supporting the photon-counting technique (XC-Thor, Direct Conversion, Danderyd, Sweden). A 40 kVp x-ray spectrum was utilized, and it was divided into two energy channels: [25.58, 40) keV and [33.20, 40) keV. The measurements falling in [33.20, 40) keV were taken as the high-energy projections, and the measurements falling in [25.58, 40) keV subtracting the measurements falling in [33.20, 40) keV were regarded as the low-energy projections. The source to detector distance was 222.76 mm, and the source to rotation center distance was 111.38 mm. The detector had 383 × 64 pixels with a size of 0.2×0.2 mm^2. A 3D volume with 384 × 384 × 32 voxels was reconstructed, and the size of each voxel was set to $0.1 \times 0.1 \times 0.1$ mm^3. The fully sampled projections were collected from the angular range of 360° with intervals of 0.6°, and the limited-angle projections were extracted from the fully sampled projection data. The FDK (ramp-filter) algorithm was adopted for reconstruction. Twenty-seven full-scans from different rats were randomly divided into a training set (21 scans, 672 slices), a validation set (3 scans, 96 slices), and a testing set (3 scans, 96 slices). In experiments, the training set was augmented to 2688 slices by three rotations with intervals of 90°. Besides, in order to eliminate the interference of noise and artifacts in the background, all the volumes were cropped to 288 × 288 × 32.

2.4. Network training and testing

In this study, the experiments were carried out with Python and TensorFlow libraries on a PC (i7-10700K 3.8 GHz CPU, 32 GB RAM, Geforce RTX 3090 24 GB GPU). The Adam algorithm with default settings was adopted to optimize the proposed TIME-Net. The initial learning rate was set to 10^{-4} and slowly decreased to 10^{-6}. The total epoch and batch size were set to 100 and 4, respectively. To exploit the global knowledge via the transformer encoder, the degraded DE-CBCT images ($512 \times 512 \times 2$ for the simulated data, $288 \times 288 \times 2$ for the real rat data) reconstructed by the FDK algorithm and the prior image ($512 \times 512 \times 1$ for the simulated data, $288 \times 288 \times 1$ for the real rat data) were directly fed into TIME-Net, rather than the image patches. The label images were reconstructed by the FDK algorithm using the full-scan data. MAE and SSIM scores calculated on the validation set were utilized to determine the optimal parameters for all models. As the statistical distributions of the two datasets were different, TIME-Net was trained on the simulated data and real rat data, respectively. During the testing, TIME-Net loaded the corresponding learned parameters.

3. Results and discussion

In this section, experimental results on simulated and real rat data are presented to assess the proposed algorithm. Four state-of-the-art methods are taken into comparison, including FDK (Feldkamp et al., 1984), SPECIAL (Hu et al., 2021), AnNet (Zhang et al., 2020c), and DD-Net++(Zhang et al., 2018). FDK is a widely used analytical reconstruction algorithm. SPECIAL is a promising DL-based artifact removal technique with single-shot projection error correction and is developed for standard single-energy CBCT. As a result, SPECIAL must deal with low- and high-energy CBCT images separately in this study. AnNet is a DECT-proprietary post-processing model that works on degraded DECT images reconstructed from dual quarter-scans. DD-Net was originally designed for sparse-view CT and here is retrained for limited-angle artifact removal. Its input and output are expanded into two channels to meet the DE-CBCT imaging problem and utilize redundant information in the energy domain. In order to distinguish it from the original DD-Net, it is termed as DD-Net++ in this study. Three metrics, MAE (Mean Absolute Error, Unit: HU), SSIM (Structural SIMilarity, Unit:

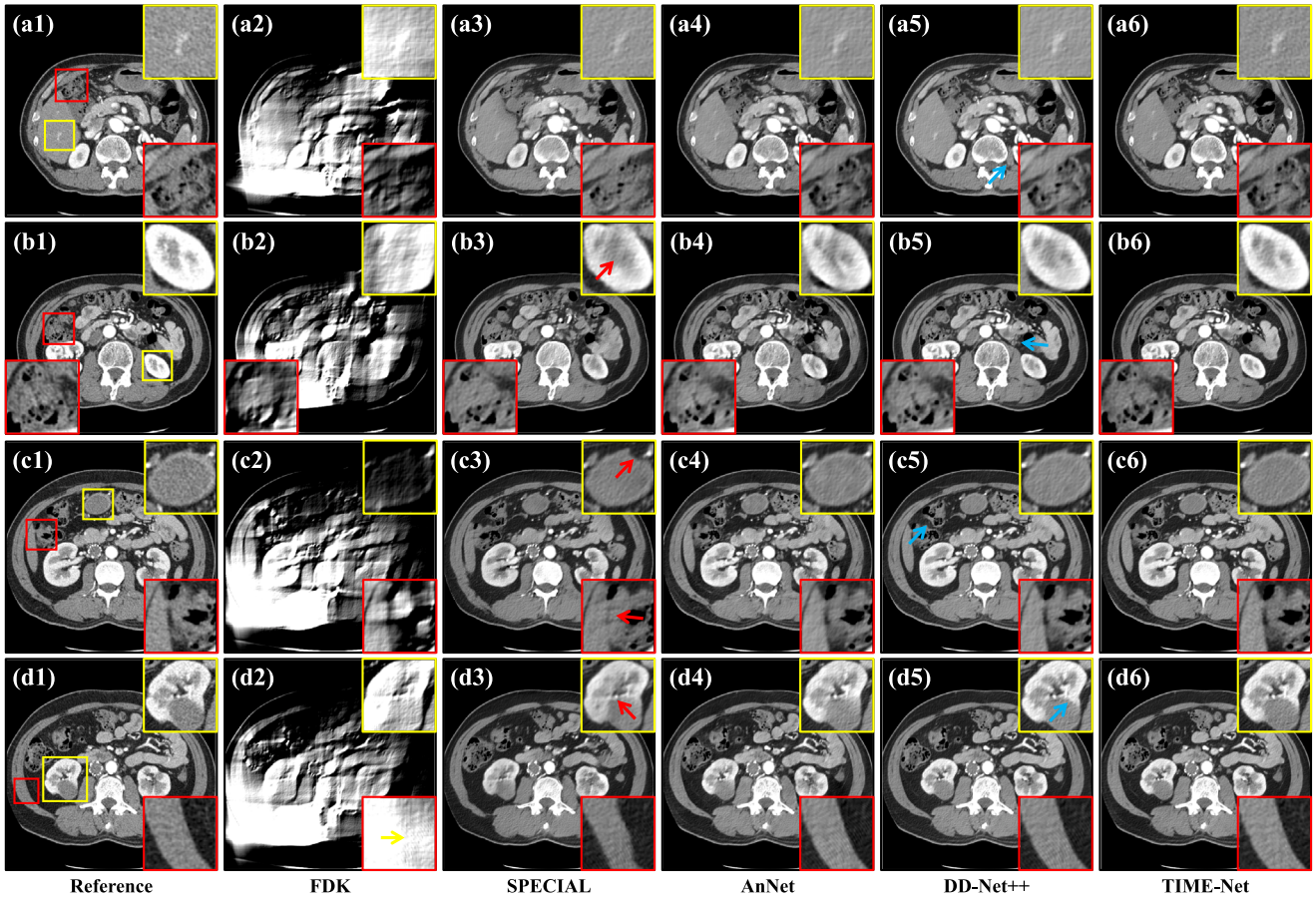


Fig. 4. The axial 100 kVp results of different methods on the simulated data. The display window is [−160, 240] HU.

%), and PSNR (Peak Signal to Noise Ratio, Unit: dB), are employed for quantitative analysis. These scores are calculated for all slices in the test set, and the average scores are presented as the final results. Moreover, two dual-energy imaging applications, VNC imaging and iodine contrast agent quantification, are also performed to investigate the clinical significance of the DE-CBCT images provided by TIME-Net.

3.1. Results on simulated data

3.1.1. Limited-angle artifact removal

Fig. 4 shows the axial 100 kVp results of different algorithms. Please note that only the single-energy (low-energy or high-energy) results are presented for a concise layout, but the proposed TIME-Net could simultaneously produce high-quality DE-CBCT images. As can be seen from Fig. 4 (a2)–(d2), the artifacts caused by incomplete sampling have severely degraded the image details and changed the CT numbers. Although SPECIAL is a state-of-the-art method for limited-angle artifact removal, it still cannot well restore the image structures without exploiting the redundancy in the energy domain, as indicated by the red arrows. Compared with the SPECIAL method, AnNet, DD-Net++, and TIME-Net demonstrate significant advantages in terms of structural fidelity, which can be found from the zoomed ROIs in Fig. 4 (a3)–(d6). Besides, the subtle features marked by the blue arrows in Fig. 4 (a5)–(d5) indicate that DD-Net++ performs more competitively than AnNet. On the whole, benefiting from the energy-domain redundancy and the global knowledge, TIME-Net outperforms other algorithms in image structure restoration, as shown in the last column of Fig. 4. And it also can be noticed that the images processed by TIME-Net have better contrast, especially in the ROIs in Fig. 4 (c6)–(d6). Furthermore, as marked by the yellow arrow in Fig. 4 (d2), FDK induces slight second artifacts in the reconstructed images. Results in the last row of Fig. 4 demonstrate that only TIME-Net can suppress these artifacts well.

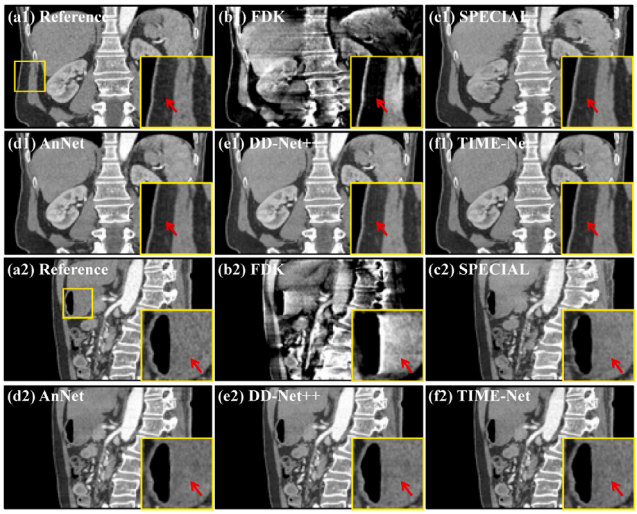


Fig. 5. The 140 kVp results of different methods on the simulated data. The coronal and sagittal images are from two different cases in the testing set. All the images are displayed in [−160, 240] HU.

The 140 kVp results of different methods are illustrated in Fig. 5. The coronal and sagittal images are selected from two different cases in the test set. It can be found that SPECIAL cannot well remove artifacts from the images reconstructed by FDK. Even though AnNet

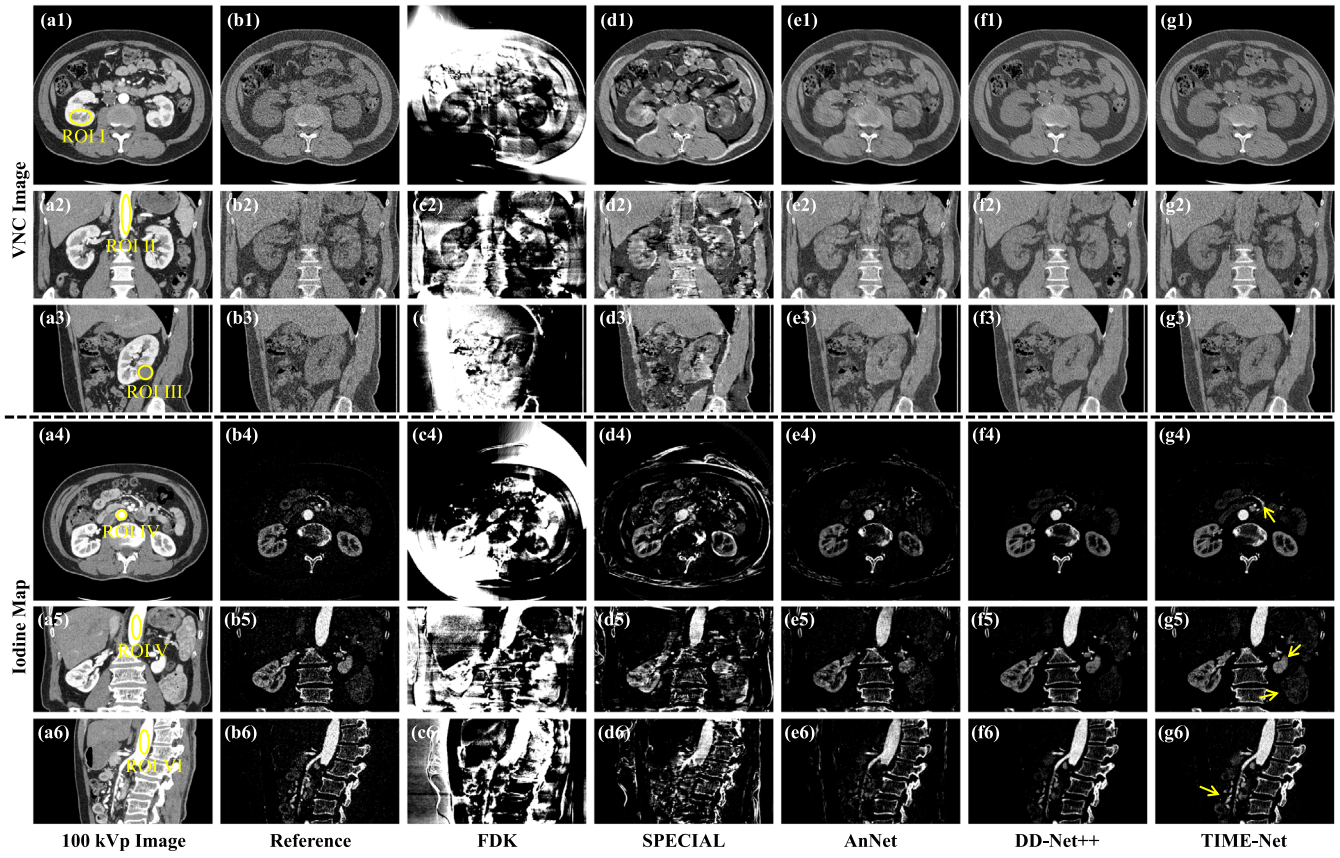


Fig. 6. VNC images and iodine maps of different methods on the simulated data. The 100 kVp images and VNC images are displayed in [−160, 240] HU, and the iodine maps are displayed in [0.1, 1.0].

Table 1
Quantitative results of different methods on the simulated data. The best scores are in bold.

Methods	MAE (HU)	SSIM (%)	PSNR (dB)
FDK	226.72	64.31	17.17
SPECIAL	32.81	90.42	31.90
AnNet	20.40	95.25	37.60
DD-Net++	18.24	96.14	38.70
TIME-Net	**16.39**	**96.94**	**39.66**

and DD-Net++ have suppressed the limited-angle artifacts, they cannot provide satisfactory performance in subtle structure preservation. In comparison, the proposed TIME-Net is promising in eliminating the artifacts and restoring the image details.

Quantitative analysis is also conducted for algorithm evaluation. The MAE, SSIM, and PSNR scores calculated on all slices in the testing set are listed in Table 1. Due to the artifacts, FDK obtains the worst scores, demonstrating that limited-angle sampling will seriously degrade the image quality. Compared with FDK, SPECIAL could greatly improve all the metrics. What is more, AnNet and DD-Net++ can provide further improvements by making use of the energy-domain redundancy. Consistent with visual assessment, TIME-Net achieves the best scores in all metrics, which profits from the synergetic contributions of three different encoders.

3.1.2. VNC imaging and iodine quantification

Dual-energy material decomposition, which exploits the energy-dependent information of DE-CBCT images to provide tissue composition information, is also performed to validate the clinical utility of the processed DE-CBCT images. Specifically, VNC imaging and iodine quantification based on the simulated data are investigated in this study. Briefly, the VNC imaging is to obtain an image in which the administrated contrast agent is artificially removed, simulating a scan without contrast agent to alleviate the need for the no-contrast-agent scanning in clinical contrast-enhanced dual-energy imaging procedure. And iodine quantification is of great clinical significance for subjectively assessing the various lesions. An image domain material decomposition method (Faby et al., 2015) is adopted to generate the VNC images and iodine maps.

VNC images and iodine maps derived from DE-CBCT images processed by different methods are presented in Fig. 6. It can be seen from Fig. 6 (c1)–(c3) that the corresponding VNC images and iodine maps are also seriously degraded due to the FDK reconstructed DE-CBCT images suffering from severe artifacts. There are some artifacts left in the VNC images and iodine maps obtained using the DE-CBCT images provided by SPECIAL and AnNet, while the results from AnNet are a bit better than SPECIAL. VNC images from DD-Net++ and TIME-Net are of few differences, while iodine maps acquired with the results of TIME-Net are more consistent with the reference, as indicated by the yellow arrows in Fig. 6 (g4)–(g6). Moreover, it can be observed that the noise in the VNC images from DD-Net++ and TIME-Net is suppressed to a certain extent, which may contribute to improving the diagnosis accuracy.

The statistical measurements (mean ±standard deviation) are performed on six representative ROIs selected from the VNC images and iodine maps to evaluate the quantification accuracy, and the results are summarized in Table 2. It can be found that FDK, SPECIAL, and AnNet cannot well preserve the quantification accuracy, either in the VNC images or in the iodine maps. In ROI I–III, DD-Net++ performs worse than TIME-Net in quantification accuracy preservation (larger mean values), although it demonstrates a little better performance in noise suppression (smaller standard deviations). As for the iodine quantification,

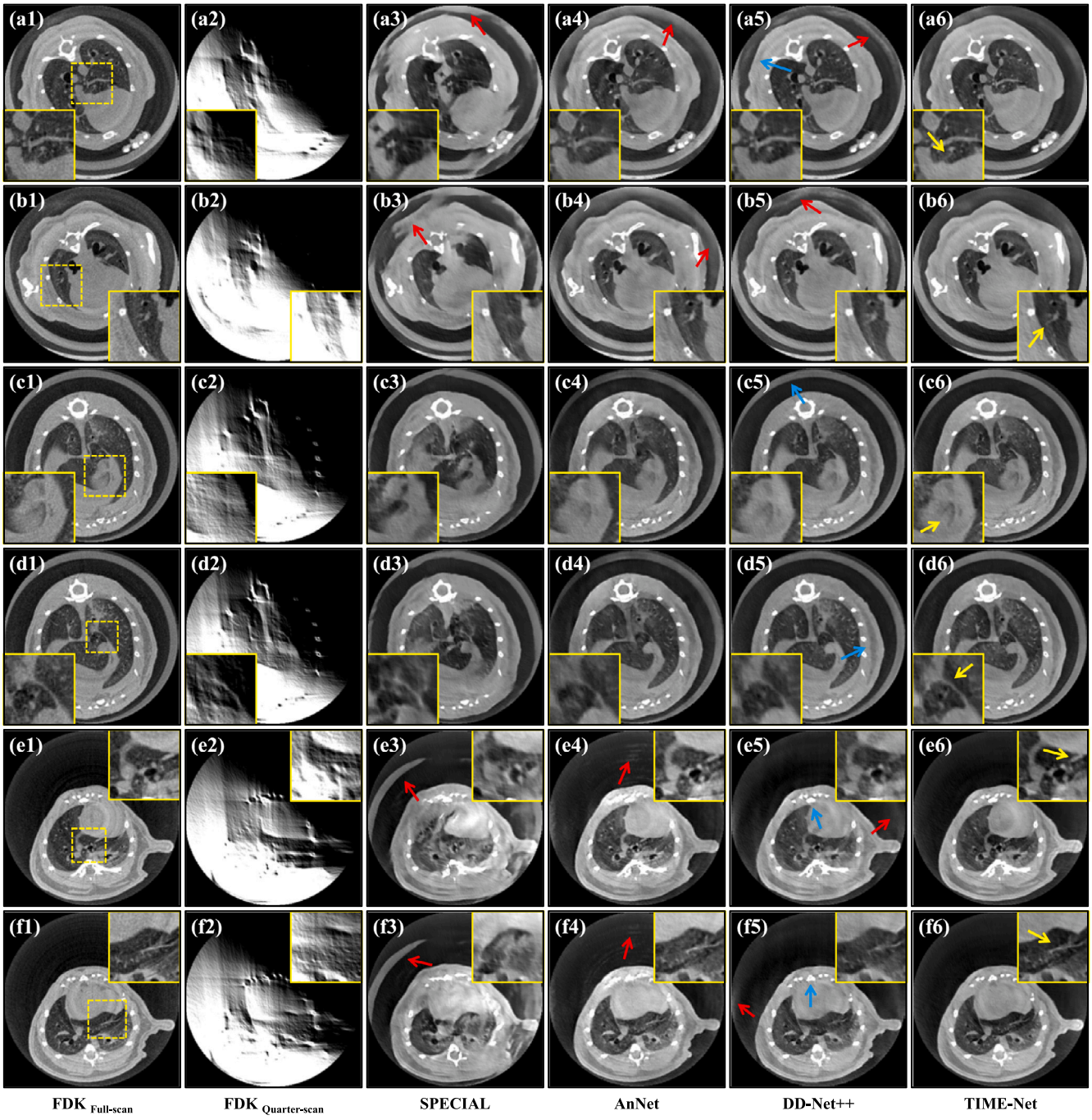


Fig. 7. The low-energy results of different methods on the real rat data. All the images are displayed in [−150, 300] HU.

Table 2
The statistical measurements (mean ± standard deviation) of VNC images and iodine maps. The ROIs are marked in Fig. 6.

Materials	ROI	Reference	FDK	SPECIAL	AnNet	DD-Net++	TIME-Net
VNC image (HU)	ROI I	28 ± 43	−54 ± 144	78 ± 56	51 ± 31	36 ± **26**	**27** ± 30
	ROI II	41 ± 47	−81 ± 166	25 ± 53	67 ± 34	44 ± **30**	**41** ± 33
	ROI III	39 ± 42	−14 ± 114	135 ± 63	46 ± **26**	52 ± **26**	**40** ± 30
Iodine map (%)	ROI IV	0.970 ± 0.138	0.525 ± 0.263	0.843 ± 0.185	0.790 ± 0.125	0.901 ± **0.097**	**0.931** ± 0.106
	ROI V	0.955 ± 0.144	1.528 ± 0.318	0.898 ± 0.139	0.820 ± 0.107	0.903 ± **0.097**	**0.905** ± 0.109
	ROI VI	0.969 ± 0.914	1.500 ± 0.421	0.862 ± 0.132	0.862 ± 0.109	**0.916** ± **0.096**	0.914 ± 0.102

DD-Net++ achieves a performance comparable to TIME-Net in ROI V and VI. However, TIME-Net shows significant superiority compared to DD-Net++ in ROI IV. On the whole, TIME-Net not only could suppress noise but also could preserve quantification accuracy, demonstrating that DE-CBCT images provided by the proposed TIME-Net are more promising in clinical applications.

Table 3
Quantitative results of different methods on the real rat data.

Methods	MAE (HU)	SSIM (%)	PSNR (dB)
FDK	226.69	44.78	12.17
SPECIAL	25.25	84.09	29.13
AnNet	16.71	91.10	33.14
DD-Net++	13.18	92.87	35.73
TIME-Net	**12.12**	**93.57**	**36.57**

Table 4
Quantitative results of ablation study. The best scores are in bold.

Image encoder	Prior encoder	Transformer encoder	MAE (HU)	SSIM (%)	PSNR (dB)
✓			17.98	96.62	39.06
✓	✓		16.85	96.71	39.39
✓		✓	16.58	96.79	39.53
✓	✓	✓	**16.39**	**96.94**	**39.66**

3.2. Results on real rat data

Fig. 7 presents the low-energy results of different methods on the real rat data. From the results reconstructed by FDK using the full-scan data, it can be observed that some ring artifacts are left in the images, especially in Fig. 7 (c1)–(f1). That is because the raw projections are not well calibrated. We have not spent much effort on ring artifact correction because the focus of this study is limited-angle artifact removal, although these ring artifacts may compromise the performance of all the involved algorithms.

The zoomed ROIs in Fig. 7 (a3)–(f3) demonstrate that the SPECIAL method almost cannot restore reliable anatomical information from the degraded DE-CBCT images. Compared with AnNet, DDNet++ performs a bit better in edge preservation, which can be found from the blue arrows in Fig. 7 (a5)–(f5). By exploring and exploiting the anatomical consistency, energy-domain redundancy and global knowledge, TIME-Net achieves the promising performance in subtle structure restoration, as pointed by the yellow arrows in Fig. 7 (a6)–(f6). Furthermore, it can be observed from the red arrows in Fig. 7 (a3)–(f6) that only the proposed TIME-Net could well deal with the interference caused by the noise and artifacts in the background.

The quantitative analysis is also performed on the real rat data, and the results are listed in Table 3. It can be found that the proposed TIME-Net obtains the beat scores in all metrics, demonstrating the promising performance of TIME-Net in artifact removal, subtle structure restoration, and reconstruction accuracy preservation.

3.3. Ablation study

In this section, an ablation study based on the simulated data is performed to evaluate the effectiveness of three encoders in TIME-Net. Table 4 lists all possible combinations of the three encoders. The image encoder is always retained as it is the main component. If an encoder is removed from the overall structure, the corresponding shallow features will no longer exist. All the models are retrained using the simulated dataset to achieve the best performance for a fair comparison.

Fig. 8 presents the 100 kVp results of the four comparison models. Because it is difficult to observe differences from images directly, the absolute error images of the ROIs are calculated and placed at the bottom right corner of the corresponding images. As indicated by the yellow arrows, the 100 kVp images produced by TIME-Net integrating all three encoders are of smaller distances to the reference than other methods.

The corresponding quantitative results are listed in Table 4. It can be found that the transformer encoder can provide bigger improvements than the prior encoder. Moreover, TIME-Net can bring further improvement and achieve promising performance in all the metrics after integrating all the encoders.

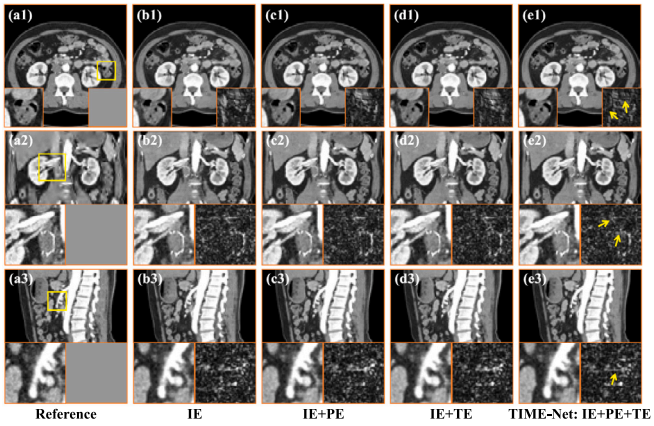


Fig. 8. The results of ablation study on the simulated data. The CT images are displayed in [−160, 240] HU, and the absolute error images are displayed in [0, 100] HU. IE: Image Encoder, PE: Prior Encoder, TE: Transformer Encoder.

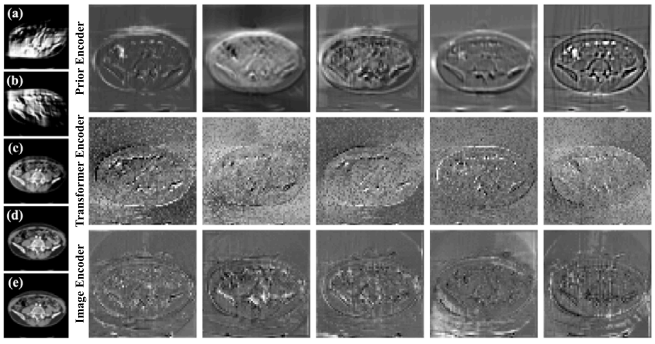


Fig. 9. Visualization of feature maps extracted by different encoders. (a) and (b) are the limited-angle DE-CBCT images. (c) is the prior image. (d) and (e) are the reference DE-CBCT images. The five columns on the right are randomly selected feature maps.

To further understand the proposed method, we randomly select and visualize some feature maps learned by different encoders, as shown in Fig. 9. It can be observed that the prior encoder contributes more information than the other two encoders in structure restoration. Features extracted by the image encoder and transformer encoder are abstract and may contain various kinds of information. However, limited by convolution operations, the image encoder has no ability to capture global information (Vaswani et al., 2017; Bello et al., 2019; Zhang et al., 2019a). The transformer encoder that can model global features of medical images makes some compensation for it (Pan et al., 2021; Zhang et al., 2021c; Wang et al., 2021a). In general, the visualized feature maps have demonstrated that the three encoders in TIME-Net play different roles, and they synergistically improve image quality.

3.4. Performance of TIME-Net under different limited-angle configurations

In real scenarios, we often need to make a trade-off between scanning range and image quality. Hence, this section investigates the performance of TIME-Net on data acquired with different limited-angle configurations. In order to exclude the influence of other factors, the simulated data were utilized in experiments. Specifically, the scan angle α in Fig. 1 was respectively set to 60°, 75°, 90°, 105° and 120°, as listed in Table 5. Then, for a fair comparison, TIME-Net was retrained using these datasets to reach the best performance. The training, validation, and test settings are the same as described in Section 2.4.

Table 5
Quantitative results of TIME-Net on the data acquired with different limited-angle configurations.

Scanning range low-energy, high-energy	MAE (HU)	SSIM (%)	PSNR (dB)
(0°, 60°], (60°, 120°]	27.29	93.41	33.24
(0°, 75°], (75°, 150°]	20.89	95.61	37.24
(0°, 90°], (90°, 180°]	16.39	96.94	39.66
(0°, 105°], (105°, 210°]	15.69	97.33	40.24
(0°, 120°], (120°, 240°]	13.98	98.11	41.58

Table 6
Quantitative results of five-fold cross-validation on the simulated and real datasets (mean ± standard deviation).

Data	Methods	MAE (HU)	SSIM (%)	PSNR (dB)
Simulated data	SPECIAL	30.40 ± 4.84	89.99 ± 2.78	32.15 ± 1.43
	AnNet	18.11 ± 1.98	94.81 ± 1.78	37.75 ± 0.94
	DD-Net++	15.71 ± 1.54	96.26 ± 1.01	39.19 ± 0.91
	TIME-Net	**13.60** ± **1.36**	**97.40** ± **0.57**	**40.06** ± **0.89**
Real rat data	SPECIAL	32.51 ± **3.28**	83.65 ± 3.96	28.28 ± 2.38
	AnNet	25.11 ± 3.36	90.50 ± 2.54	31.41 ± **1.93**
	DD-Net++	18.58 ± 3.33	92.13 ± 2.16	33.71 ± 2.48
	TIME-Net	**17.64** ± 3.30	**92.93** ± **1.74**	**34.13** ± 2.23

Table 5 quantifies the performance of TIME-Net under different limited-angle configurations. It can be observed that the distance between the DE-CBCT images provided by TIME-Net and the reference images decreases as the scanning range increases.

To further understand the influence of scan range, we also present the visual results, as shown in Fig. 10. It can be found that the proposed TIME-Net could well remove the wedge artifacts when the scan angle $\alpha = 60°$, although the image structure suffers from some deformation (red arrows). When the scanning angle increases to 75°, the image edges are well preserved (blue arrows), but the image details remain to be further improved, as shown by the zoomed ROIs in Fig. 10 (b1) and (d1). Results of TIME-Net demonstrate good consistency with the reference images when the scanning angle is not less than 90°, and it is hard to observe the slight improvement of image quality through subjective assessment. Hence, two complementary quarter-scans may be a good point to balance the reconstruction accuracy and scanning range.

3.5. Five-fold cross-validation

To investigate the robustness and generalizability of the involved DL-based methods, the five-fold cross-validation was performed. Both the simulated dataset and real rat dataset were randomly divided into five folds. In each validation, four folds were used for training, and the remaining one was utilized for testing. Each fold of the simulated dataset contained five patients. The first two folds of the real rat dataset had six rats, and the last three folds included five rats.

Quantitative results of five-fold cross-validation on the simulated and real data are summarized in Table 6. Compared with other methods, the proposed TIME-Net achieves better scores and smaller standard deviations on the simulated data, which demonstrates the promising performance and better robustness and generalizability of TIME-Net. On the real rat data, TIME-Net also obtains the smaller MAE and higher SSIM and PSNR compared to competing methods. Although SPECIAL and AnNet respectively achieve the smallest standard deviations in MAE and PSNR, these two methods need to be further improved. Moreover, compared with TIME-Net, these two methods have not significantly shown better robustness and generalizability.

3.6. Discussion

Although the proposed TIME-Net has demonstrated encouraging performance for limited-angle artifact removal in DE-CBCT, some issues still remain to be addressed. First, the proposed TIME-Net is a supervised method, and it is trained to minimize the distance between the network output and the ground truth (high-quality DE-CBCT images). However, limited by current technical bottlenecks, there may be noise and artifacts left in the ground truth images, which may lower the performance of the proposed algorithm in real scenarios. In this study, it can be found that the noise and artifacts in the real rat data obviously compromise the performance of all the involved methods to varying degrees. Second, TIME-Net is a data-driven model. The statistical distributions of datasets collected with different devices or configurations may be different. The experiments find that the parameters trained with simulated data cannot be directly applied to real rat data. Therefore, TIME-Net needs to be retrained when processing new data. How to transfer the trained model to other datasets also is a challenging problem. Third, due to the lack of real DE-CBCT data from patients, we were not able to evaluate the proposed algorithm on clinical data. Hence, we released the code of this study (https://github.com/YKZhangSEU/TIME-Net) in the hope that researchers having access to raw DE-CBCT data can freely evaluate the proposed TIME-Net. And the proposed method also can be used for DECT data since TIME-Net itself does not depend on the imaging geometry. Finally, even though the potential clinical significance of DE-CBCT imaging has been investigated through phantom, animal, and few patient experiments, it has not been routine practice for clinical diagnosis and treatment. There is still a gap between objective image quality improvement and its clinical utility. The close collaboration with radiologists and vendors, which could push the algorithm study into clinical deployment, is very important in future work.

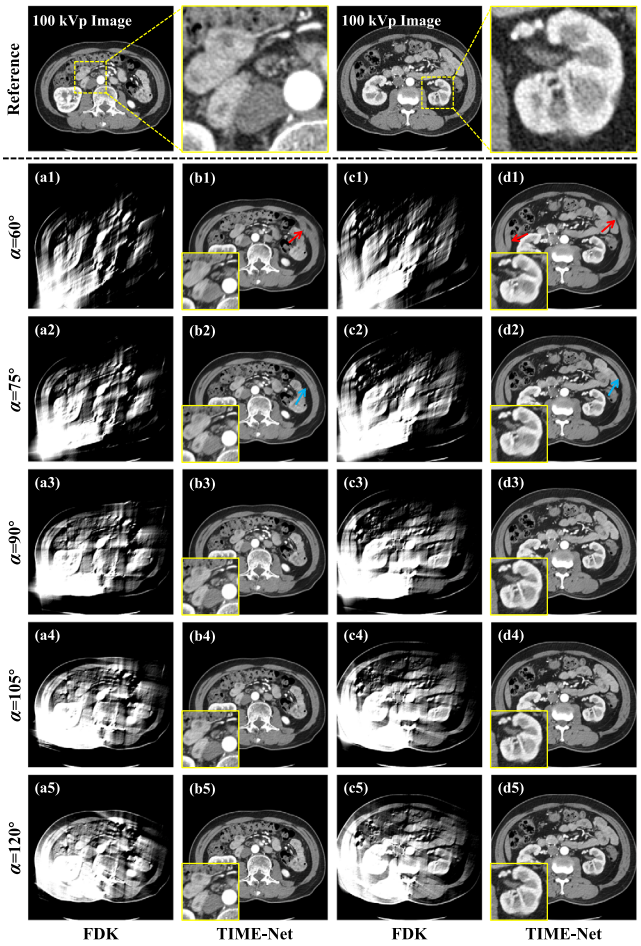


Fig. 10. Visual results of TIME-Net on the data acquired with different limited-angle configurations. All the images are displayed in [−160, 240] HU.

4. Conclusion

DE-CBCT is a promising imaging technology and has the potential to be deployed in the clinic to improve dentistry, radiotherapy, interventions, and so on. Meanwhile, real-time 3D volume imaging is very important in some clinical applications, such as image-guided radiation therapy. Hence, this paper develops TIME-Net to produce high-quality DE-CBCT images from the degraded images reconstructed with two complementary quarter-scans, reducing the radiation dose and shortening the acquisition time. The ablation study has verified the effectiveness of the three encoders involved in TIME-Net. And the experimental results on the simulated and real rat data have demonstrated the promising performance of TIME-Net in artifact removal, subtle structure restoration, and accuracy preservation. The potential utility in clinical applications of the DE-CBCT images processed by TIME-Net has also been assessed by VNC imaging and iodine quantification.

CRediT authorship contribution statement

Yikun Zhang: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft. **Dianlin Hu:** Software, Validation, Writing – review & editing. **Zhihong Yan:** Investigation. **Qingxian Zhao:** Data curation. **Guotao Quan:** Writing – review & editing. **Shouhua Luo:** Investigation, Resources, Writing – review & editing. **Yi Zhang:** Writing – review & editing. **Yang Chen:** Supervision, Project administration, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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